Course notes

Reminders: conditioning and independence

If $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. A tree branch carries a conditional probability; probability a path is the product of its branches. For a partition $A_1, \ldots, A_r$, $P(B) = \sum_i P(A_i \cap B)$; if all the $P(A_i) > 0$, it is written $\sum_i P(A_i) P_{A_i}(B)$. Two events are independent when $P(A \cap B) = P(A)P(B)$. For $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. Not to be confused $P_A(B)$ and $P_B(A)$.

Example. $P(A) = 0{,}4$, $P_A(B) = 0{,}7$, $P_{\overline{A}}(B) = 0{,}2$. Then $P(B) = 0{,}4 \times 0{,}7 + 0{,}6 \times 0{,}2 = 0{,}4$ and $P_B(A) = \frac{0{,}28}{0{,}4} = 0{,}7$.

Bernoulli and binomial

A Bernoulli event has two issues, success of probability p and probability failure $1-p$. The indicator variable B is 1 if successful and 0 if not: $E(B)=p$, $V(B)=p(1-p)$. A schematic of Bernoulli is a rehearsal of n the same and independent tests. X of success follows binomial distribution $\mathcal{B}(n,p)$.

Law and parameters

For $k \in \{0,\dots,n\}$,

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}.$$

$$E(X)=np, \qquad V(X)=np(1-p), \qquad \sigma(X)=\sqrt{np(1-p)}.$$

Demonstration of the law. Each path comprising k success a probability $p^k(1-p)^{n-k}$ by independence. There is $\binom{n}{k}$ choice of success positions. These paths are incompatible; add up their probabilities.

Cumulative events

$$P(X\le k)=\sum_{j=0}^{k}P(X=j), \qquad P(X\ge k)=1-P(X\le k-1).$$

$$P(a\le X\le b)=P(X\le b)-P(X\le a-1).$$

For at least one success, $P(X\ge 1)=1-(1-p)^n$. The addition of « at least k » is "strictly less than k ».

Example. For $X\sim\mathcal{B}(10,0{,}2)$, $E(X)=2$ and $V(X)=1{,}6$. $P(X=2)=45(0{,}2)^2(0{,}8)^8\simeq 0{,}3020$ and $P(X\ge 1)=1-0{,}8^{10}\simeq 0{,}8926$. Expectation is neither a certain result nor necessarily the most likely.

Threshold and interval

To look for the smallest n such as $1-(1-p)^n\ge q$, a loop on n After the logarithm, we can solve $(1-p)^n\le 1-q$ then round in the right direction. To find the smallest integer k such that $P(X>k)\le\alpha$, increase k until $P(X\le k)\ge 1-\alpha$, then check the previous row. We can also look for an interval $[a,b]$ containing at least one prescribed probability, without assume that it is unique.

Independent tests, beyond two issues

For independent experiments, the probability of a list of results is the product of probabilities. The universe is the Cartesian product of the universes of each experience. By example, for two independent throws of a balanced dice, the universe contains the 36 pairs (i,j), $1\le i,j\le 6$, each probability $\frac{1}{36}$. The probability that the sum is 5 $\frac{4}{36}=\frac{1}{9}$, because couples in favour are $(1,4),(2,3),(3,2),(4,1)$. In a Bernoulli diagram, we code success by 1 and Failed by 0: the universe is $\{0,1\}^n$. Independence concerns the whole succession; a simple equality between two observed probabilities is not sufficient to prove it.

Build the model before calculation

13.1 From the table to the tree

First use: S36.

Example Solved · Change denominator

Of the 1,000 objects, 600 come from workshop A and 400 from B. A produces 12 defective objects and B in product 20. By noting D the defect, $P(A \cap D) = 12/1000$, $P_A(D) = 12/600 = 0{,}02$ and $P_D(A) = 12/32 = 0.375$. The same number 12 occurs, but the reference world changes.

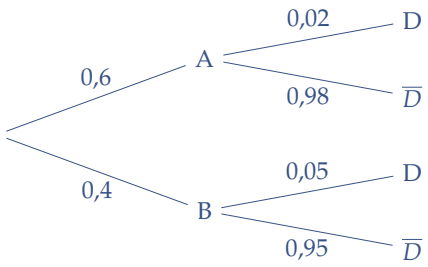


Figure 13.1. Read path A and then D: $0.6 \times 0{,}02 = 0.012$, or 12 items out of 1,000.

A1 · Immediate application Complete the table with two lines A/B and two columns D/no D. Calculate $P(D)$ Are the defect and the workshop independent?

13.2 Find a branch and reverse conditioning

First use: S38.

Ownership · Partition and Total Probability

If A and $\overline{A}$ form a partition, $P(D) = P(A)P_A(D) + P(\overline{A})P_{\overline{A}}(D)$, when the conditional probabilities This result comes from the separate meeting of the two paths leading to D.

Example Solved · Missing Data

We know. $P(A) = 0{,}7$, $P_A(D) = 0{,}1$ and $P(D) = 0{,}13$. Poisons $x = P_{\overline{A}}(D)$. Equation 0 ,13 $= 0{,}7 \times 0{,}1 + 0{,}3x$ gives $x = 0{,}2$. The result belongs to $[0;1]$, which is a necessary control.

A2 · Partial guidance With $P(A) = 0{,}4$, $P_A(D) = 0.2$ and $P(D) = 0.14$, find $P_{\overline{A}}(D)$ then $P_D(A)$.

A3 · Refuter "If 90% of patients have a positive test, then 90% of positive people are Build a picture of 1,000 people, with 100 sick, 90 positive among the sick. and 90 positives among non-malads, then calculate the two proportions.

Four characteristics to be checked

First use: S39 and S41.

Definition · Schematic of Bernoulli

Repeat a fixed number n independent tests, with two exits and the same probability of success p. The number X of success follows the law $\mathcal{B}(n;p)$: $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$ for $0\le k\le n$.

Example Solved · Count Paths

For four probability tests 1 /2, achieving exactly two successes corresponds to six paths, each probability 1 /16. Thus $P(X=2)=6/16$. Independence makes it possible to multiply the probabilities of a path; the binomial coefficient counts the investments of success.

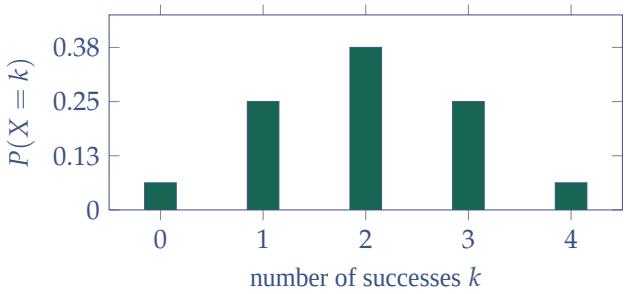


Figure 13.2. Law $\mathcal{B}(4;1/2)$: explain the symmetry and check the sum of the heights.

A4 · Accept or refuse Say in which cases the binomial model is justified: ten independent throws 1 coin; 10 prints without discount in an urn containing five red balls and five blues ; number of launches before the first success. Justify each answer.

13.4 Translate Words into Integer Events

First use: S41.

Example resolved · A loose border of a unit

For X integer, "strictly more than 4" means $X\ge 5$ and its complement is $X\le 4$. "At least 4" means $X\ge 4$ and its complement is $X\le 3$. Finally, $P(2\le X\le 5)=P(X\le 5)-P(X\le 1)$.

A5 · Immediate application For $X\sim\mathcal{B}(8;0{,}3)$, write using cumulative probabilities events: at least 3 hits; strictly less than 3; between 2 and 5 included. Calculate exactly $P(X=0)$.

13.5 Determine a probability rather than a number of tests

First use: S42.

Example resolved · Two tests

For two independent probability trials p, the probability of at least one success is $1-(1-p)^2$. To make it equal to 3/4 is to $(1-p)^2 = 1/4$. Like $1 - p \geq 0$, we keep $1 - p = 1/2$, so p=1/2. The negative sign root would be incompatible with p in [0;1].

A6 · Choice of strategy Three trials are independent, probability of success p. We want the probability of at least two successes of 0 ,9. Express this probability based on p, demonstrate the uniqueness of a solution in $[0;1]$ and frame between two hundredths consecutive.

A first step: In order to be unique, study a function on the $[0;1]$.

Wait a minute at the courthouse. Explain why a probability tree does not involve independence: probabilities on the second level may depend on the first result.

Review postponed · S50 and then S73 Solve a threshold $1 - (1-p)^n \geq c$ by logarithm, then compare exact probabilities and concentration limits.

Research activity

Component control · 40 min.

A component is chosen at random in a production from two workshops. A Event "the component comes from the first workshop", $B = \overline{A}$ and D The event "the component is We're giving. $P(A) = 0{,}6$, $P_A(D) = 0{,}02$ and $P_B(D) = 0{,}05$. The results will be compared Rounded to 10^{-4}.

1. Represent the situation by a weighted tree, then calculate $P(D)$.

2. The chosen component is defective. Calculate the probability that it comes from the first workshop.

3. Twenty components are now selected independently, each following the previous model.
   Note X the number of defective components among the twenty. Justify the law followed by X and specify its parameters.

4. Calculate $P(X = 0)$, $P(X \geq 2)$ and $E(X)$.

5. If the 20 components were taken from the same batch, what hypothesis should the model be?
   Please explain why.